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Course: Software Testing and Quality Assurance Laboratory	Code: BIT26PE11
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Assignment 6: Perform Black Box testing using automated testing tools on an application. Testing Points to be covered – data driven wizard	

1. OBJECTIVE

- To design and implement automated **Black Box test cases** that verify the "Add Room" functionality using a **Data-Driven Wizard**.
- To decouple test logic from test data by utilizing an external data source.
- To verify the system's ability to process multiple sequential data entries without manual intervention.

2. TEST ENVIRONMENT & CONFIGURATION

- **Operating System:** Windows 11 (XAMPP Environment).
- **Automation Tool:** Selenium WebDriver for Python.
- **Data Source:** External CSV file (room_data.csv) containing 5 test records.
- **Browser:** Google Chrome managed via WebDriverManager.

3. IMPLEMENTATION DETAILS (DATA-DRIVEN LOGIC)

Black Box testing focuses on the functional requirements of the application. In this assignment, we use a "Wizard" pattern to automate the complex, multi-input "Add Room" form.

3.1 Data Source Preparation (room_data.csv)

The "Wizard" relies on an external file to provide input values for each iteration.

- **Test Data:** Contains 5 unique records (testroom1 to testroom5) with varying area, price, and capacity values.

room_data.csv

name,area,price,quantity,adult,children,desc
testroom1,200,500,5,2,1,Standard automated test room 1
testroom2,250,600,4,2,2,Standard automated test room 2
testroom3,300,750,3,3,1,Standard automated test room 3

```
testroom4,350,900,2,2,3,Standard automated test room 4
testroom5,400,1100,2,4,2,Standard automated test room 5
```

3.2 Wizard Automation Logic

- **Iterative Execution:** The script implements a for loop that reads each row of the CSV and executes the "Add Room" workflow.
- **Form Mapping:** The script maps CSV columns directly to the name, area, price, quantity, adult, children, and desc fields found in the add-room modal.
- **Dynamic Selection:** The automation identifies and interacts with checkboxes for **Features** and **Facilities**, simulating a full user experience.

4. TEST EXECUTION SCRIPT

```
import csv
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.service import Service
from webdriver_manager.chrome import ChromeDriverManager

service = Service(ChromeDriverManager().install())
driver = webdriver.Chrome(service=service)

def run_room_wizard_assignment6(csv_file):
    try:
        driver.get("http://localhost/hbwebsite/admin/index.php")
        driver.maximize_window()
        driver.find_element(By.NAME, "admin_name").send_keys("####")
        driver.find_element(By.NAME, "admin_pass").send_keys("####")
        driver.find_element(By.NAME, "login").click()
        time.sleep(2)

        driver.get("http://localhost/hbwebsite/admin/rooms.php")
        time.sleep(2)

        with open(csv_file, 'r') as file:
            reader = csv.DictReader(file)
            for row in reader:
                print(f"Wizard is processing: {row['name']}")
                driver.find_element(By.XPATH, "//button[@data-bs-target='#add-room']").click()
                time.sleep(1)

                driver.find_element(By.NAME, "name").send_keys(row['name'])
                driver.find_element(By.NAME, "area").send_keys(row['area'])
                driver.find_element(By.NAME, "price").send_keys(row['price'])
                driver.find_element(By.NAME, "quantity").send_keys(row['quantity'])
                driver.find_element(By.NAME, "adult").send_keys(row['adult'])
                driver.find_element(By.NAME, "children").send_keys(row['children'])
```

```

driver.find_element(By.NAME, "desc").send_keys(row['desc'])

features = driver.find_elements(By.NAME, "features")
if features:
    features[0].click()

driver.find_element(By.XPATH, "//button[contains(text(), 'SUBMIT')]").click()
print(f"Record {row['name']} added successfully.")

time.sleep(3)
driver.refresh()

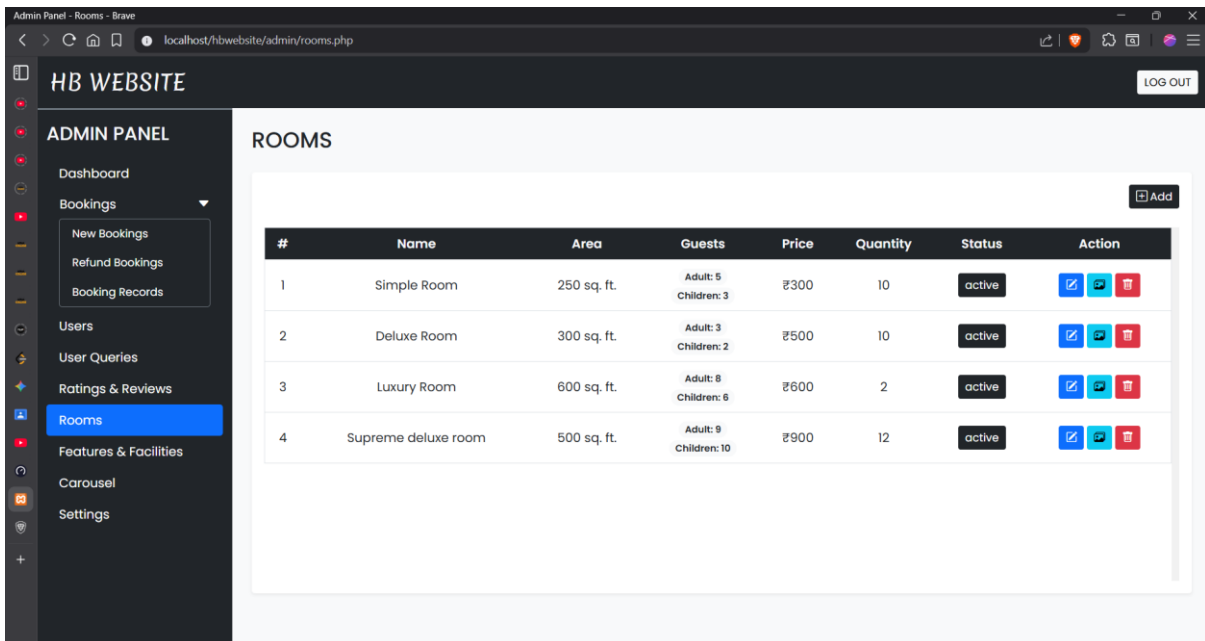
except Exception as e:
    print(f"Automation Error: {e}")

if __name__ == "__main__":
    run_room_wizard_assignment6("room_data.csv")
    driver.quit()

```

5. TEST RESULTS & OBSERVATIONS

5.1 Pre-Execution State (Before)



#	Name	Area	Guests	Price	Quantity	Status	Action
1	Simple Room	250 sq. ft.	Adult: 5 Children: 3	₹300	10	active	[Edit] [Add] [Delete]
2	Deluxe Room	300 sq. ft.	Adult: 3 Children: 2	₹500	10	active	[Edit] [Add] [Delete]
3	Luxury Room	600 sq. ft.	Adult: 8 Children: 6	₹600	2	active	[Edit] [Add] [Delete]
4	Supreme deluxe room	500 sq. ft.	Adult: 9 Children: 10	₹900	12	active	[Edit] [Add] [Delete]



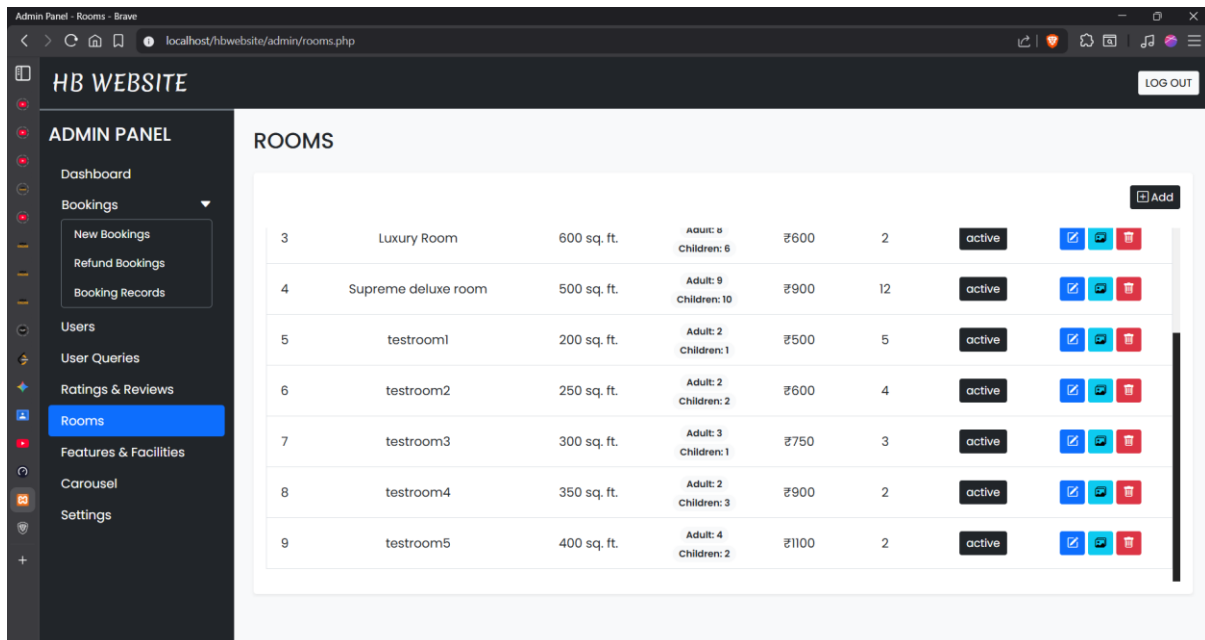
```

room_data.csv > data
1 name,area,price,quantity,adult,children,desc
2 testroom1,200,500,5,2,1,Standard automated test room 1
3 testroom2,250,600,4,2,2,Standard automated test room 2
4 testroom3,300,750,3,3,1,Standard automated test room 3
5 testroom4,350,900,2,2,3,Standard automated test room 4
6 testroom5,400,1100,2,4,2,Standard automated test room 5

```

5.2 Post-Execution State (After)

```
[Running] python -u "c:\xampp\htdocs\hbwebsite\assignment6.py"
Wizard is processing: testroom1
Record testroom1 added successfully.
Wizard is processing: testroom2
Record testroom2 added successfully.
Wizard is processing: testroom3
Record testroom3 added successfully.
Wizard is processing: testroom4
Record testroom4 added successfully.
Wizard is processing: testroom5
Record testroom5 added successfully.
```



The screenshot shows the 'HB WEBSITE' Admin Panel. The left sidebar contains the 'ADMIN PANEL' menu with options: Dashboard, Bookings (with sub-options: New Bookings, Refund Bookings, Booking Records), Users, User Queries, Ratings & Reviews, Rooms (highlighted), Features & Facilities, Carousel, and Settings. The main content area is titled 'ROOMS' and displays a table of room data. An 'Add' button is in the top right corner of the table.

ID	Room Name	Area (sq. ft.)	Adults	Children	Price (₹)	Capacity	Status	Actions
3	Luxury Room	600 sq. ft.	Adult: 8	Children: 6	₹600	2	active	[Edit] [View] [Delete]
4	Supreme deluxe room	500 sq. ft.	Adult: 9	Children: 10	₹900	12	active	[Edit] [View] [Delete]
5	testroom1	200 sq. ft.	Adult: 2	Children: 1	₹500	5	active	[Edit] [View] [Delete]
6	testroom2	250 sq. ft.	Adult: 2	Children: 2	₹600	4	active	[Edit] [View] [Delete]
7	testroom3	300 sq. ft.	Adult: 3	Children: 1	₹750	3	active	[Edit] [View] [Delete]
8	testroom4	350 sq. ft.	Adult: 2	Children: 3	₹900	2	active	[Edit] [View] [Delete]
9	testroom5	400 sq. ft.	Adult: 4	Children: 2	₹1100	2	active	[Edit] [View] [Delete]

6. CONCLUSION

The use of an automated **Data-Driven Wizard** significantly increases test coverage while reducing the need for redundant script writing. By automating the "Add Room" module, we verified that the Admin Panel can handle high-volume data entries and correctly interface with the underlying MySQL database.